

IN3060/INM460 Computer Vision Coursework report

- **Student name, ID and cohort:** Bhuvanesi Barua (240067790) - PG
- **Google Drive folder:**
<https://drive.google.com/drive/folders/1Kpd5yJWOP55z6wMKD2PhEOx4mpHIN-9-?usp=sharing>

Data preprocessing

- Loaded dataset from image directory and label file; resized all images to 128×128×3 for consistency.
- Converted labels into integer format representing 4 classes (Child, Young, Middle, Senior).
- Split dataset into training (9310 samples) and validation (3990 samples) to monitor model performance.
- Maintained class distribution across splits (e.g., majority class = Young, minority classes balanced across sets).
- For SVM and MLP:
 - Converted images to grayscale to reduce dimensionality.
 - Extracted HOG features (8×8 pixels per cell, 2×2 cells per block, 9 orientations) for structured representation.
- For CNN:
 - Used raw RGB images to allow automatic feature learning.
- Saved processed datasets (X_train, X_val, y_train, y_val, HOG features) to avoid recomputation and reduce runtime.
- Ensured consistent preprocessing between training, validation, and test sets to avoid data leakage.

Implemented models

- Model no. 1: HOG + SVM
 - HOG features used as input representation.
 - Kernel type (linear, rbf (Radial Basis Function)) and regularisation parameter C (1, 10) evaluated using cross-validation
 - Best configuration selected based on highest mean accuracy.
 - Best configuration: rbf kernel, C=10
- Model no. 2: HOG + MLP
 - HOG features used as input to a feedforward neural network.
 - Hidden layer sizes ((64,), (128,), (128,64)) and solvers (adam, sgd (Stochastic Gradient Descent)) evaluated.
 - Best architecture and optimiser selected using cross-validation performance.
 - Best configuration: hidden layer (128,), solver adam
- Model no. 3: CNN
 - Raw images used directly as input.
 - Architecture composed of three convolutional layers followed by fully connected layers.
 - Trained for up to 10 epochs with batch size 32.
 - EarlyStopping applied (patience=2, monitored on validation loss) to retain best-performing model.

Quantitative results

	Accuracy	Precision	Recall	F1-score
1.HOG + SVM	0.775	0.771	0.775	0.772
2.HOG + MLP	0.738	0.736	0.738	0.736
3.CNN	0.78	0.781	0.78	0.775